

Laborator 1 - SO

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Comenzi laborator

```
• lucian-rus@fedora:~/Desktop/dev/school/uni-courses/so/tema2$ echo ceva > file.txt
• lucian-rus@fedora:~/Desktop/dev/school/uni-courses/so/tema2$ cat file.txt
ceva
• lucian-rus@fedora:~/Desktop/dev/school/uni-courses/so/tema2$ cat file.txt | gzip > file.txt.gz
• lucian-rus@fedora:~/Desktop/dev/school/uni-courses/so/tema2$ zcat file.txt.gz
ceva
• lucian-rus@fedora:~/Desktop/dev/school/uni-courses/so/tema2$ cat < file.txt
ceva
• lucian-rus@fedora:~/Desktop/dev/school/uni-courses/so/tema2$ cat file.txt
ceva
• lucian-rus@fedora:~/Desktop/dev/school/uni-courses/so/tema2$ echo "linia 2" >> file.txt
• lucian-rus@fedora:~/Desktop/dev/school/uni-courses/so/tema2$ cat file.txt
ceva
linia 2
• lucian-rus@fedora:~/Desktop/dev/school/uni-courses/so/tema2$ echo -n "linia 3" >> file.txt
• lucian-rus@fedora:~/Desktop/dev/school/uni-courses/so/tema2$ echo "continua" >> file.txt
• lucian-rus@fedora:~/Desktop/dev/school/uni-courses/so/tema2$ echo "linia 4" >> file.txt
• lucian-rus@fedora:~/Desktop/dev/school/uni-courses/so/tema2$ cat file.txt
ceva
linia 2
linia 3continua
linia 4
• lucian-rus@fedora:~/Desktop/dev/school/uni-courses/so/tema2$ tar -cvf extra.tar extra/
extra/
extra/file1
extra/file2
extra/extra-dir/
extra/extra-dir/file3
• lucian-rus@fedora:~/Desktop/dev/school/uni-courses/so/tema2$ tar -xvf extra.tar -C dup
• lucian-rus@fedora:~/Desktop/dev/school/uni-courses/so/tema2$ ls -R dup/
dup:/
extra

dup/extra:
extra-dir  file1  file2

dup/extra/extra-dir:
file3
• lucian-rus@fedora:~/Desktop/dev/school/uni-courses/so/tema2$ tar -c extra/ | gzip > extra.tgz
```

```
• lucian-rus@fedora:~/Desktop/dev/school/uni-courses/so/tema2$ tar zcf extra.tar.gz extra/
• lucian-rus@fedora:~/Desktop/dev/school/uni-courses/so/tema2$ tar zxf extra.tar.gz -C dup/
• lucian-rus@fedora:~/Desktop/dev/school/uni-courses/so/tema2$ zip -r extra.zip extra/*
adding: extra/extra-dir/ (stored 0%)
adding: extra/extra-dir/file3 (stored 0%)
adding: extra/file1 (stored 0%)
adding: extra/file2 (stored 0%)
• lucian-rus@fedora:~/Desktop/dev/school/uni-courses/so/tema2$ unzip extra.zip
Archive:  extra.zip
replace extra/extra-dir/file3? [y]es, [n]o, [A]ll, [N]one, [r]ename: A
extracting: extra/extra-dir/file3
extracting: extra/file1
extracting: extra/file2
• lucian-rus@fedora:~/Desktop/dev/school/uni-courses/so/tema2$ ls -l
commands.txt
dup
extra
extra.tar
extra.tar.gz
extra.tgz
extra.zip
file.txt
file.txt.gz
• lucian-rus@fedora:~/Desktop/dev/school/uni-courses/so/tema2$ man grep && grep -n grep
```

```

GREP(1) User Commands GREP(1)

NAME
  grep - print lines that match patterns

SYNOPSIS
  grep [OPTION]... PATTERNS [FILE]...
  grep [OPTION]... -e PATTERNS ... [FILE]...
  grep [OPTION]... -f PATTERN-FILE ... [FILE]...

DESCRIPTION
  grep searches for patterns in each FILE. In the synopsis's first form, which is used if no -e or -f options are present, the first operand PATTERNS is one or more patterns separated by newline characters, and grep prints each line that matches a pattern. Typically PATTERNS should be quoted when grep is used in a shell command.

  A FILE of "-" stands for standard input. If no FILE is given, recursive searches examine the working directory, and nonrecursive searches read standard input.

OPTIONS
  Generic Program Information
  --help Output a usage message and exit.

  -V, --version
        Output the version number of grep and exit.

  Pattern Syntax
  -E, --extended-regexp
        Interpret PATTERNS as extended regular expressions (ERES, see below).

  -F, --fixed-strings
        Interpret PATTERNS as fixed strings, not regular expressions.

  -G, --basic-regexp
        Interpret PATTERNS as basic regular expressions (BRES, see below). This is the default.

Manual page grep(1) line 1 (press h for help or q to quit)

```

```

• lucian-rus@fedora:~/Desktop/dev/school/uni-courses/so/tema2$ cat file.txt
words keep on coming
on and on and ON here
and onto here also
• lucian-rus@fedora:~/Desktop/dev/school/uni-courses/so/tema2$ sed 's/on/off/' file.txt
words keep off coming
off and on and ON here
and offto here also
• lucian-rus@fedora:~/Desktop/dev/school/uni-courses/so/tema2$ sed 's/on /off /' file.txt
words keep off coming
off and on and ON here
and onto here also
• lucian-rus@fedora:~/Desktop/dev/school/uni-courses/so/tema2$ sed 's/on /off /g' file.txt
words keep off coming
off and off and ON here
and onto here also
• lucian-rus@fedora:~/Desktop/dev/school/uni-courses/so/tema2$ sed 's/on /off /gi' file.txt
words keep off coming
off and off and off here
and onto here also
• lucian-rus@fedora:~/Desktop/dev/school/uni-courses/so/tema2$ sed 's/here/(&)/g' file.txt
words keep on coming
on and on and ON (here)
and onto (here) also
○ lucian-rus@fedora:~/Desktop/dev/school/uni-courses/so/tema2$ 

```

```

• lucian-rus@fedora:~/Desktop/dev/school/unl-courses/so/tema2$ sed 's/\([a-zA-Z0-9]\)[a-zA-Z0-9]*\)/\([a-zA-Z0-9]\)[a-zA-Z0-9]*\)/\2 \1/' file.txt
xt
keep words on coming
and on on and OM here
onto and here also
• lucian-rus@fedora:~/Desktop/dev/school/unl-courses/so/tema2$ sed 's/\([^\ ]*\) \([^\ ]*\)/\2 \1/' file.txt
keep words on coming
and on on and OM here
onto and here also
• lucian-rus@fedora:~/Desktop/dev/school/unl-courses/so/tema2$ sed -i 's/\([^\ ]*\) \([^\ ]*\)/\2 \1/' file.txt
• lucian-rus@fedora:~/Desktop/dev/school/unl-courses/so/tema2$ cat file.txt
keep words on coming
and on on and OM here
onto and here also
• lucian-rus@fedora:~/Desktop/dev/school/unl-courses/so/tema2$ 

```

```

• lucian-rus@fedora:~/Desktop/dev/school/unl-courses/so/tema2$ cat cmmdc.sh
1: #!/bin/bash
2:
3: cmmdc () {
4:
5: a=$1
6: b=$2
7:
8: echo "> $a $b"
9:
10: if [ $b -eq 0 ]; then
11: return $a
12: fi
13: c=$(( $a % $b ))
14: cmmdc $b $c
15: return $?
16: }
17: echo "parametrii sunt: $0, primul: $1, al doilea: $2"
18:
19: if [ $1 -lt $2 ]; then
20: echo "ordine incorecta"
21: exit 1
22: fi
23:
24: cmmcl $1 $2
25: echo "c.m.m.d.c. este: $?"
• lucian-rus@fedora:~/Desktop/dev/school/unl-courses/so/tema2$ sed -i 's/[0-9]\(1,\)/ /' cmmdc.sh
• lucian-rus@fedora:~/Desktop/dev/school/unl-courses/so/tema2$ ./cmmdc.sh
parametrii sunt: , primul: , al doilea:
ordine incorecta
• lucian-rus@fedora:~/Desktop/dev/school/unl-courses/so/tema2$ ./cmmdc.sh 273 26
parametrii sunt: 273 26, primul: 273, al doilea: 26
./cmmdc.sh: line 35: cmmcl: command not found
c.m.m.d.c. este: 127
• lucian-rus@fedora:~/Desktop/dev/school/unl-courses/so/tema2$ 

```

Teme

```
ex-tema > $ ex2.sh
1  #!/bin/bash
2
3  # pune arg1 in n -> comanda folosita './ex2.sh <rang>'
4  n=$1
5
6  # daca avem 1 sau 0, iesim
7  if [ $n -le 0 ]; then
8      echo 0
9  elif [ $n -eq 1 ]; then
10     echo 1
11 else
12     previous=0
13     current=1
14
15     # calculam fibonnaci
16     for ((i=2; i<=n; i++)); do
17         next=$((previous + current))
18         previous=$current
19         current=$next
20     done
21
22     # daca am ajuns la final, printeaza numarul
23     echo $current
24 fi
25
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS MEMORY XRTOS

• lucian-rus@fedora: ~/Desktop/dev/school/uni-courses/so/tema2/ex-tema\$./ex2.sh 13
233
✂ lucian-rus@fedora: ~/Desktop/dev/school/uni-courses/so/tema2/ex-tema\$

Figure 1: Exercitiul 2

Determinarea termenului Fibonnaci de rang n . Operatii efectuate:

- daca $n = 0$, printam numarul
- daca $n = 1$, printam numarul
- calculam termenul Fibonnaci prin utilizarea unui **for** loop

```
ex-tema > $ ex3.sh
1  #!/bin/bash
2  sudo find / -type f | grep "\.conf$" > ex3.output
3
4  # tine numele dir-ului precedent
5  previous=""
6  total_sum=0
7
8  # du-te prin output de la comanda find
9  for file in `cat ex3.output`; do
10     # echo $file
11     dir=$(sudo dirname $file)
12     # sa in calcul doar daca previous si dir sunt diferite -> avem fisier nou
13     if [ "$dir" != "$previous" ]; then
14         # echo $dir
15         previous=$dir
16         # suprima potentiale probleme cauzate de fisiere cu spatiu in nume si verifica daca numele e gol sau nu
17         size=$(sudo du -s "$dir" 2>/dev/null | cut -f1)
18         if [ -n "$size" ]; then
19             total_sum=$((total_sum + size))
20         fi
21     fi
22 done
23
24 echo "total sum: $total_sum KB"

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  MEMORY  XRTOS

* lucian-rus@fedora: ~/Desktop/dev/school/unl-courses/so/tema2/ex-tema$ ./ex3.sh
find: '/run/user/1000/doc': Permission denied
find: '/run/user/1000/gvfs': Permission denied
total sum: 588404 KB
~ lucian-rus@fedora: ~/Desktop/dev/school/unl-courses/so/tema2/ex-tema$
```

Figure 2: Exercitiul 3

Determinarea sumei dimensiunilor tuturor directoarelor care contin fisiere de configuratii (fisiere terminate in **.config**). Operatii efectuate:

- listam toate fisierele care se termina in **.conf** folosind **find** si **grep** si salvam continutul intr-un fisier
- iteram prin continutul fisierului rezultat si extragem directorul in care se afla fisierele utilizand comanda **dirname**
- verificam daca directorul curent e diferit de cel precedent; acest lucru ne ajuta sa evitam fisierele duplicat
- daca fisierul nu e duplicat, determinam dimensiunea lui folosind comanda **du**; additional, facem o verificare pentru preventia erorilor generate de directoarele cu spatii in nume
- cumulam dimensiunea in variabila **total_sum** si o afisam la final

```

ex-tema > $ ex4.sh
1  #!/bin/bash
2
3  # creeaza fisier
4  touch ex4.output
5
6  # itereaza si indeplineste conditiile din tema
7  for ((i=1; i<=100; i++)); do
8      echo "linia $i: continut" >> ex4.output
9  done
10

```

Figure 3: Exercițiul 4

```

ex-tema > $ ex5.sh
1  # ruleaza tot cu -i ca sa inlocuim content-ul fisierului
2  sed -i 's/continut/content/' ex4.output
3  sed -i 's/linia/line/' ex4.output
4  sed -i 's/:/ -/' ex4.output
5
6  cat ex4.output

```

Figure 4: Exercițiul 5

Generarea unui fisier si alterarea continutului sau. Operatii facute:

- cream un fisier gol folosind comanda **touch**
- cream continutul cerut de exercitiu cu ajutorul unui loop **for**; fiecare linie contine indexul la care se afla; numerotarea incepe de la 1
- inlocuim continutul in interiorul fisierului utilizand comanda **sed** si flag-ul **-i**
- listam continutul fisierului folosind **cat**

Explicatia acopera atat exercitiul 4 cat si exercitiul 5.